PROBLEM STATEMENT

With the rapid advancement and widespread adoption of Artificial Intelligence across industries such as IT, Finance, Healthcare, Manufacturing, Retail, Marketing, and Education, there is growing concern about how AI technologies are transforming the workforce. Organizations are witnessing significant changes in job roles, employment status, salary structures, and productivity levels. However, due to the large volume and complexity of employment data, traditional data processing systems struggle to efficiently analyze and extract meaningful patterns about AI's impact on employees. There is a critical need for a scalable, distributed computing solution that can process thousands of employee records and generate actionable insights about workforce transformation.

This project aims to address this challenge by leveraging the Hadoop ecosystem, including HDFS for distributed storage, MapReduce for batch processing, and Hive for analytical querying. The primary goal is to analyze key metrics such as job status distribution (Replaced, Modified, Unchanged), salary changes before and after AI adoption, productivity variations based on AI adoption levels, automation risk assessment across industries, and upskilling requirements by job role. The insights derived from this analysis will help organizations understand which industries and job roles are most affected by AI, identify skill gaps requiring upskilling, and develop strategic workforce planning to navigate the AI-driven economy effectively.